

Hoeffding-Based Safety Probability Proposal for Arbitrary Trajectories

Objective

The goal is to validate whether an arbitrary candidate trajectory is safe with a statistical lower bound on the probability of safety. Instead of proving that a trajectory converges to a nominal reference, the proposal estimates the probability that a candidate trajectory satisfies safety constraints under sampled uncertainty.

Proposed Validation Artifact

The proposed validation samples can be stored in:

`papers/report/arbitrary_trajectory_safety_validation_placeholder.csv`.

Each row stores one sampled rollout or perturbation test for a candidate trajectory.

Arbitrary Candidate Trajectory

Let an arbitrary candidate trajectory be a finite horizon sequence of states and controls.

$$\tau = \{(x_0, u_0), (x_1, u_1), \dots, (x_T, u_T)\}$$

Plug in the UAV planning interpretation.

$$x_t = \begin{bmatrix} p_{x,t} \\ p_{y,t} \\ p_{z,t} \end{bmatrix}, \quad u_t = \begin{bmatrix} v_{x,t} \\ v_{y,t} \\ v_{z,t} \end{bmatrix}$$

Final result.

τ can be any generated, sampled, RRT, LLM, or hand-designed trajectory.

Safety Set

The trajectory is safe if every state remains inside the workspace and outside all obstacle safety margins.

$$x_t \in \mathcal{X}_{\text{safe}}, \quad u_t \in \mathcal{U}, \quad t = 0, \dots, T.$$

Define the safe set as

$$\mathcal{X}_{\text{safe}} = \mathcal{X} \setminus \bigcup_{\ell=1}^{N_{\text{obs}}} \mathcal{O}_{\ell}^{r_{\text{safe}}},$$

where $\mathcal{O}_\ell^{r_{\text{safe}}}$ is obstacle ℓ expanded by safety radius r_{safe} . Plug in the current planner-style safety margin.

$$r_{\text{safe}} = 0.40 \text{ m}$$

Final result.

$$x_t \in \mathcal{X}_{\text{safe}} \text{ means the trajectory remains at least } 0.40 \text{ m away from obstacles.}$$

Safety Indicator

For a sampled rollout $X_i(\tau)$ of trajectory τ , define an indicator variable.

$$I(X_i(\tau)) := \begin{cases} 1, & \text{if } x_t \in \mathcal{X}_{\text{safe}}, u_t \in \mathcal{U}, \forall t = 0, \dots, T, \\ 0, & \text{otherwise.} \end{cases}$$

In words, $I(X_i(\tau)) = 1$ means the sampled rollout of the arbitrary trajectory is safe. Plug in one example outcome.

$$I(X_i(\tau)) = 1 \quad \Rightarrow \quad \min_t d(x_t, \mathcal{O}) \geq 0.40 \text{ m}$$

Final result.

$$I(X_i(\tau)) \in [0, 1] \text{ is a Bernoulli safety test.}$$

True Safety Probability

The true probability that trajectory τ is safe is

$$\mu_\tau := \mathbb{P}[I(X_i(\tau)) = 1].$$

This probability is taken over sampled uncertainty such as obstacle noise, tracking error, perception error, wind disturbance, or initial-state perturbation. Plug in the interpretation.

$$\mu_\tau = \mathbb{P}[\tau \text{ remains collision-free and constraint-satisfying}]$$

Final result.

$$\mu_\tau \text{ is the safety probability of the arbitrary trajectory.}$$

Empirical Safety Rate

Given p independent sampled rollouts of the same arbitrary trajectory, estimate μ_τ by

$$\tilde{\mu}_\tau := \frac{1}{p} \sum_{j=1}^p I(X_j(\tau)).$$

Plug in an example validation batch with $p = 100$ rollouts and 96 safe rollouts.

$$\tilde{\mu}_\tau = \frac{96}{100} = 0.96$$

Final result.

$$\tilde{\mu}_\tau = 0.96$$

Hoeffding Inequality

Let $I(X_j(\tau))$, $j = 1, \dots, p$, be p iid random variables with

$$0 \leq I(X_j(\tau)) \leq 1.$$

Hoeffding's inequality gives

$$\mathbb{P} [|\tilde{\mu}_\tau - \mu_\tau| \geq \epsilon_h] \leq 2 \exp(-2p\epsilon_h^2).$$

Define the confidence level

$$\delta_h := 2 \exp(-2p\epsilon_h^2).$$

Final result.

$$\boxed{\mathbb{P} [|\tilde{\mu}_\tau - \mu_\tau| < \epsilon_h] \geq 1 - \delta_h}$$

Lower Bound on Safety Probability

Rearrange Hoeffding's inequality to obtain a lower confidence bound.

$$\mu_\tau \geq \tilde{\mu}_\tau - \epsilon_h$$

with confidence at least $1 - \delta_h$. Solve for ϵ_h from

$$\delta_h = 2 \exp(-2p\epsilon_h^2).$$

This gives

$$\epsilon_h = \sqrt{\frac{\ln(2/\delta_h)}{2p}}.$$

Final result.

$$\boxed{\mu_\tau \geq \tilde{\mu}_\tau - \sqrt{\frac{\ln(2/\delta_h)}{2p}}}$$

Numerical Example

Use $p = 100$ sampled rollouts and confidence parameter $\delta_h = 0.05$.

$$\epsilon_h = \sqrt{\frac{\ln(2/0.05)}{2(100)}}$$

$$\epsilon_h = \sqrt{\frac{\ln(40)}{200}} = \sqrt{\frac{3.688879}{200}} = 0.135812$$

Using $\tilde{\mu}_\tau = 0.96$, the safety probability lower bound is

$$\mu_\tau \geq 0.96 - 0.135812 = 0.824188.$$

Final result.

$$\boxed{\mu_\tau \geq 0.824188 \text{ with confidence at least } 0.95.}$$

Acceptance Rule

Choose a required safety probability μ_{crit} . Accept the arbitrary trajectory if

$$\mu_{\text{crit}} \leq \tilde{\mu}_\tau - \epsilon_h.$$

Plug in $\mu_{\text{crit}} = 0.80$.

$$0.80 \leq 0.96 - 0.135812 = 0.824188$$

Final result.

τ is accepted as statistically safe.

Required Number of Rollouts

If the desired empirical gap is ϵ_h , solve Hoeffding's inequality for p .

$$\delta_h = 2 \exp(-2p\epsilon_h^2)$$

$$p \geq \frac{\ln(2/\delta_h)}{2\epsilon_h^2}.$$

Plug in $\delta_h = 0.05$ and $\epsilon_h = 0.05$.

$$p \geq \frac{\ln(40)}{2(0.05)^2} = \frac{3.688879}{0.005} = 737.7758$$

Final result.

$p \geq 738$ sampled rollouts are required.

Trajectory-Level Safety Score

For ranking many candidate trajectories, define the lower confidence bound score

$$J_{\text{safe}}(\tau) := \tilde{\mu}_\tau - \sqrt{\frac{\ln(2/\delta_h)}{2p}}.$$

The selected trajectory is

$$\tau^\star = \arg \max_{\tau \in \mathcal{T}_{\text{cand}}} J_{\text{safe}}(\tau)$$

subject to

$$J_{\text{safe}}(\tau) \geq \mu_{\text{crit}}.$$

Final result.

Choose the trajectory with the largest certified lower bound on safety.

Closed-Loop Control Integration Ideas

Idea 1: Safety Filter Before Execution

Generate candidate trajectories and validate each trajectory with the Hoeffding lower bound.

$$J_{\text{safe}}(\tau) \geq \mu_{\text{crit}}$$

Only trajectories satisfying the bound are sent to the controller. Final result.

Unsafe or statistically uncertain trajectories are rejected before flight.

Idea 2: Receding-Horizon Revalidation

At each control update, validate the next short-horizon trajectory segment.

$$\tau_{t:t+H} = \{x_t, u_t, \dots, x_{t+H}, u_{t+H}\}.$$

Accept the segment if

$$J_{\text{safe}}(\tau_{t:t+H}) \geq \mu_{\text{crit}}.$$

Final result.

The safety probability is refreshed online as perception and state estimates change.

Idea 3: Backup Controller Trigger

If the lower bound drops below the safety threshold, switch to a verified backup controller.

$$J_{\text{safe}}(\tau) < \mu_{\text{crit}} \quad \Rightarrow \quad u = u_{\text{backup}}(x).$$

The backup controller can hover, brake, land, or track a locally safe waypoint. Final result.

Closed-loop control uses the probability bound as a runtime safety trigger.

Idea 4: Risk-Aware Cost for MPC

Include the lower-bound safety score in the MPC objective.

$$\min_{\tau} \sum_{t=0}^H \ell(x_t, u_t) \lambda_{\text{risk}} [1 - J_{\text{safe}}(\tau)]$$

subject to

$$J_{\text{safe}}(\tau) \geq \mu_{\text{crit}}.$$

Final result.

The controller prefers trajectories with higher certified safety probability.

Summary

The proposed method turns arbitrary trajectory safety into a Bernoulli validation problem. For each candidate trajectory, sampled rollouts produce an empirical safety rate $\tilde{\mu}_{\tau}$. Hoeffding's inequality converts this empirical rate into a lower confidence bound:

$$\mu_{\tau} \geq \tilde{\mu}_{\tau} - \sqrt{\frac{\ln(2/\delta_h)}{2p}}$$

This bound can be used as a trajectory verifier, a planner ranking score, a receding-horizon safety check, or a trigger for a backup controller.